JOHN FAN

johnfan.dev@gmail.com | 281-509-4259 | linkedin.com/in/johnwfan | github.com/johnwfan | johnwfan.com

EDUCATION

Rice University Houston, TX

Bachelor of Science in Computer Science | GPA: 3.79 *Expected May 2028*

- **Relevant Coursework**: Algorithmic Thinking (Algorithms & Discrete Math), Program Design (Java), Computer Organization (C), Matrix Analysis, Probability & Statistics

EXPERIENCE

Screenz.ai May 2026 – Aug. 2026

Software Engineer Intern *Remote*

- Built and shipped an AI cost-attribution system and admin COGS dashboard, instrumenting ~20 call sites across 6 provider AI/LLM APIs to attribute customer spend and forecast monthly costs.
- Rebuilt a prospect PDF pipeline with stricter competitor matching and a 12-check GEO/AEO site audit; parallelized processing to cut report latency by up to ~10 seconds.
- Fixed onboarding data loss across OAuth, email signup, and early drop-off by persisting qualification data sooner and backfilling affected accounts with an idempotent PostgreSQL migration; validated with 20 Jest and 4 Playwright tests.
- Fixed admin impersonation across browser restarts by rebuilding sessionStorage from identity cookies and NextAuth JWTs, restoring admin controls and clearing stale state on logout.

iCode Aug. 2024 – Mar. 2025

Software & Robotics Instructor *Pearland, TX*

- Taught Python, JavaScript, HTML/CSS, and VEX robotics to classes of 15–25 students, guiding students through debugging, design, build, and test cycles.

SELECTED PROJECTS

Scuttle.gg (League of Legends) | *Next.js, TypeScript, MongoDB, Redis, FastAPI* Nov. 2025 – Present

- Built and deployed a Riot API analytics platform supporting 11 regions with ranked stats, champion mastery, match history, performance trends, and full-lobby analysis.
- Implemented a typed Riot API client with regional routing, input validation, retry/backoff, and Upstash Redis caching with workload-specific TTLs to reduce duplicate requests and rate-limit pressure.
- Built a FastAPI draft-analysis pipeline on 14.8K Master+ matches, evaluating logistic regression and CatBoost with rolling-forward validation across patches; identified limited predictive signal and exposed it as a low-confidence draft edge rather than a win probability.

Flow (HackRice 16) | *Next.js, TypeScript, TimescaleDB, Docker, Vultr* Sep. 2026

- Co-built a real-time focus coach combining webcam pulse and breathing readings (Presage SDK) with active-window tracking to classify focus states and trigger interventions; camera frames stay on-device.
- Designed and built the full Next.js frontend, including the design system and the live-session and history views for focus state, pulse, breathing, and session insights.
- Set up TimescaleDB for time-series storage and analytics and deployed the Dockerized application to Vultr behind a custom domain with HTTPS.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, Java, SQL, C, HTML/CSS
Frameworks & Libraries: Next.js, React, Node.js, FastAPI, Prisma, scikit-learn, CatBoost
Tools & Databases: PostgreSQL, MongoDB, Redis, TimescaleDB, Docker, Linux, Vercel, Vultr, Git, Playwright